

Office Attendance System

Administrator Operations Manual

DOCUMENT TYPE	Administrator Manual
SYSTEM	ESP32 Fingerprint Attendance — Multi-Branch
AUDIENCE	Branch administrators
VERSION	2.0

Table of Contents

1. Getting Started

- 1.1 What to Expect When the Device Arrives
- 1.2 What Has Already Been Done For You
- 1.3 Your Responsibilities as Administrator

2. First-Time WiFi Setup

- 2.1 Powering On the Device
- 2.2 Connecting the Device to Office WiFi
- 2.3 Verifying the Connection

3. Enrolling the Master Finger

- 3.1 What the Master Finger Does
- 3.2 How to Enroll the Master Finger
- 3.3 Verifying the Enrollment

4. Enrolling Employees

- 4.1 How Enrollment Works
- 4.2 Registering a New Employee
- 4.3 What Happens on the Device
- 4.4 Verifying a Successful Enrollment

5. Day-to-Day Operations

- 5.1 How Employees Use the Device
- 5.2 Monitoring Attendance in Real Time
- 5.3 Reading the Attendance Record
- 5.4 Understanding Status Values

6. Ongoing Admin Tasks

- 6.1 Adding a New Employee
- 6.2 Removing an Employee
- 6.3 Replacing a Lost or Changed Fingerprint
- 6.4 Changing the Office WiFi Network
- 6.5 Emergency Portal Access (No Master Finger)
- 6.6 Reviewing the Monthly Email Report
- 6.7 Understanding the Monthly Archive
- 6.8 Correcting a Wrong Record
- 6.9 Checking for Missed Check-Outs

7. Troubleshooting

- 7.1 Device LED Colour Guide
- 7.2 Common Problems and Solutions

Getting Started

What you need to know before you begin

1.1 What to Expect When the Device Arrives

When you receive your attendance device it will be fully assembled and pre-configured with your branch's details. The only things you need to do before it is ready for daily use are: connect it to your office WiFi network, and enrol fingerprints for your employees. Everything else has already been done for you.

NOTE

The device has no battery. It must be plugged into a power source whenever it is in use. Keep it plugged in at all times during the work day.

1.2 What Has Already Been Done For You

- The backend system has been deployed and is fully operational.
- Your branch has been registered in the central system.
- A Google Sheet for your branch has been created and linked to the system.
- The device firmware has been flashed and is ready to use.
- The device has been configured with your branch ID.

1.3 Your Responsibilities as Administrator

As branch administrator, your ongoing responsibilities are:

- **First-time setup:** Connect the device to WiFi and enrol the master finger.
- **Employee management:** Enrol new employees, remove leavers, and replace fingerprints when needed.
- **WiFi maintenance:** Reconfigure the WiFi connection if the office network changes.
- **Record keeping:** Monitor and correct attendance records in Google Sheets as needed.
- **Monthly reports:** Review the automated monthly summary email sent to your inbox.

First-Time WiFi Setup

Connecting the device to your office network

2.1 Powering On the Device

Plug the device into a wall socket or USB power source using the cable provided. Once powered on, the fingerprint sensor LED will begin glowing — this confirms the device is running. On first boot the device has no WiFi credentials saved and will enter setup mode automatically. The sensor LED will pulse **purple** to indicate it is waiting to be configured.

2.2 Connecting the Device to Office WiFi

- 1 On your phone or laptop, open your WiFi settings and look for a network named **Attendance-Setup**.
- 2 Connect to **Attendance-Setup**. The password is **setup1234**.
- 3 Once connected, open a web browser and go to: **http://192.168.4.1**. A setup page will appear.
- 4 Enter your office WiFi name (SSID) and password into the form.
- 5 Click **Save & Connect**. The device will save the credentials and restart automatically.
- 6 Reconnect your phone or laptop to your normal office WiFi after this step.

NOTE

If the setup page does not appear automatically after connecting to Attendance-Setup, type <http://192.168.4.1> manually into your browser's address bar.

WARNING

Do not share the Attendance-Setup password (setup1234) with employees. Anyone who knows it and has physical access to the device could reconfigure it.

2.3 Verifying the Connection

After the device restarts, check the sensor LED:

LED State	Meaning
Steady blue	Connected to WiFi successfully. Ready to use.
Steady purple	WiFi connection failed. Check your password and try again from Step 1.

TIP

If the LED stays purple after restarting, the most common cause is a wrong WiFi password. Repeat Section 2.2 and re-enter the credentials carefully.

Enrolling the Master Finger

Setting up the administrator fingerprint

3.1 What the Master Finger Does

The master finger is a special administrator fingerprint enrolled on the device. Its primary purpose is to trigger the WiFi setup mode — useful when you need to change the office WiFi network. When the device recognises the master finger it flashes yellow five times and launches the WiFi setup portal.

NOTE

The master finger does not record attendance. Scanning it will never create an entry in the Attendance sheet. It is purely an admin tool. Enrol it using your own fingerprint, or the fingerprint of whoever is responsible for maintaining the device.

3.2 How to Enroll the Master Finger

Enrollment is managed from the Google Sheet for your branch. Make sure the device is powered on and its LED is steady blue before proceeding.

- 1 Open your branch spreadsheet in Google Sheets.
- 2 Click on the tab labelled **Enrollment** at the bottom of the sheet.
- 3 Click on the first empty row below the header row.
- 4 Fill in the columns as shown in the table below.
- 5 Within about 10 seconds the device LED will turn solid yellow. This means it has picked up the enrollment request and is waiting for a finger.
- 6 Place your finger firmly on the sensor and hold it still until the LED changes.
- 7 Lift your finger when the LED goes off briefly, then place the same finger again for the second scan.
- 8 If successful the LED flashes green. The Status column will update to **registered** within a few seconds.

Field	Description
FingerID	Leave blank. The device fills this in after enrollment.
UniqueID	Enter any short identifier, e.g. MASTER-01.
Name	Enter a label, e.g. Admin Master.
Role	Type exactly: master
Command	Type exactly: register

Status	Leave blank.
Note	Leave blank.
CreatedAt	Optional — you may enter today's date for your records.

WARNING

Do not enrol the master finger with Role set to **employee**. If you do, it will record attendance instead of launching the WiFi portal.

3.3 Verifying the Enrollment

After enrollment, confirm the following in the Enrollment sheet:

- The **FingerID** column now contains a number (e.g. 1).
- The **Status** column reads **registered**.
- The **Role** column reads **master**.

To confirm it works, place the master finger on the sensor. The LED should flash yellow five times then turn purple — the WiFi portal is now running. Since you do not need to reconfigure WiFi right now, simply unplug and replug the device to restart it normally. The blue LED will return.

TIP

Test the master finger immediately after enrollment so you know it works before you ever need it in an emergency.

Enrolling Employees

Registering fingerprints for your team

4.1 How Enrollment Works

All enrollment is managed from the **Enrollment** tab in your branch's Google Sheet. You add a row with the employee's details and a **register** command, and the device detects it automatically within about 10 seconds. It then guides the employee through two finger placements, stores the fingerprint, and updates the sheet with the result.

TIP

Have the employee standing next to the device before you add the row to the sheet, so they are ready when the yellow LED appears.

4.2 Registering a New Employee

- 1 Open the branch spreadsheet in Google Sheets and click the **Enrollment** tab.
- 2 Click on the first empty row below the last entry.
- 3 Fill in the columns as shown in the table below.
- 4 Within about 10 seconds the device LED will turn solid yellow, indicating it is ready for the first scan.
- 5 Ask the employee to place their finger firmly and flatly on the sensor and hold it still.
- 6 The LED will go off briefly. Ask the employee to lift their finger and wait for the LED to turn yellow again.
- 7 Ask the employee to place the same finger on the sensor again for the second scan.
- 8 If successful the LED flashes green and the Status column updates to **registered** within a few seconds.

Field	Description
FingerID	Leave blank. The device fills this in automatically.
UniqueID	The employee's Staff ID (e.g. EMP-042). Must be unique per employee.
Name	The employee's full name (e.g. Kwame Mensah).
Role	Type exactly: employee
Command	Type exactly: register
Status	Leave blank.
Note	Leave blank.

CreatedAt	Optional — you may enter today's date for your records.
-----------	---

WARNING

Each employee must use the same finger every time they scan — both for enrollment and for all future attendance scans. Choose a finger with a clear, undamaged print. Index fingers usually work best.

4.3 What Happens on the Device During Enrollment

LED State	Meaning	Action
Solid yellow (first)	Waiting for first finger placement.	Place the employee's finger on the sensor.
Solid yellow (second)	First scan done — waiting for same finger again.	Ask the employee to place the same finger again.
Green flash	Enrollment successful.	Confirm Status = registered in the sheet.
Red flash	Enrollment failed.	Check the Note column for an error code. Retry per Section 7.2.

4.4 Verifying a Successful Enrollment

Confirm the following before the employee walks away:

- The **FingerID** column contains a number.
- The **Status** column reads **registered**.

Ask the employee to do a test scan immediately. Their Time-In should appear as a new row in the Attendance sheet within a few seconds. If no row appears, see Chapter 7.

Day-to-Day Operations

How the system runs once set up

5.1 How Employees Use the Device

Employee use is simple: place their registered finger on the sensor when arriving at the office, and place the same finger when leaving. Nothing else is required.

- A **green flash** means the scan was accepted.
- A **red flash** means the fingerprint was not recognised. The employee should try again.
- The first scan of the day is recorded as **Time-In**.
- Every scan after that updates **Time-Out**. The last scan of the day is the final departure time.
- Once a scan is accepted the employee must wait at least **60 seconds** before scanning again.

5.2 Monitoring Attendance in Real Time

Open the branch spreadsheet in Google Sheets at any time to see live attendance. New rows appear in the Attendance sheet within a few seconds of each scan. You do not need to refresh the page.

TIP

Bookmark the branch spreadsheet URL for quick access during the work day.

5.3 Reading the Attendance Record

Each row in the Attendance sheet represents one employee's record for one day:

Field	Description
Date	Date of the record, in YYYY-MM-DD format.
Employee Name	The employee's full name.
Staff ID	The employee's unique staff ID.
Time-In	Time of the first scan of the day (HH:MM format).
Time-Out	Time of the last scan of the day (HH:MM format). Blank if the employee has not yet scanned out.
Status	Computed automatically. See Section 5.4 below.
Column G	Used internally by the system to prevent duplicate records. Do not delete, edit, or hide it.

5.4 Understanding Status Values

The Status column is computed automatically every time an employee scans, and is finalised by the nightly job at 11:30 PM.

Status	Meaning
On-Time	Employee arrived and left within the allowed grace window.
Late	Employee's Time-In was after the allowed start grace period.
Left-Early	Employee's Time-Out was before the allowed end grace period.
Late Left-Early	Employee arrived late and also left early.
No Check-Out	Employee scanned in but did not scan out before the nightly job ran.
Late No Check-Out	Employee arrived late and also did not scan out.
Absent	No scan was recorded for the entire day. Inserted automatically by the nightly job.

Ongoing Admin Tasks

Everything you may need to do after first setup

6.1 Adding a New Employee

Follow the same process as Chapter 4. In brief:

- 1 Open the branch spreadsheet and go to the Enrollment tab.
- 2 Add a new row with the employee's Staff ID, Name, Role (**employee**), and Command (**register**).
- 3 Have the employee stand at the device and wait for the yellow LED.
- 4 Guide them through the two finger placements.
- 5 Confirm the Status reads **registered** before the employee leaves.

6.2 Removing an Employee

When an employee leaves the organisation, remove their fingerprint from the device so it can no longer be used.

- 1 Open the Enrollment tab of the branch spreadsheet.
- 2 Find the employee's row and note their **FingerID** number.
- 3 In that row, change the **Command** cell to **delete** and clear the **Status** cell.
- 4 The device will pick up the command within 10 seconds and delete the fingerprint. The Status column will update to **deleted**.

NOTE

Deleting a fingerprint does not remove the employee's historical attendance records. All past records remain in the Attendance sheet and in archive tabs.

WARNING

Once deleted, the employee cannot scan in again until re-enrolled. Only delete employees who have genuinely left.

6.3 Replacing a Lost or Changed Fingerprint

If an employee can no longer be recognised — due to a cut, injury, or switching to a different finger — delete their current fingerprint and register a new one.

- 1 Follow Section 6.2 to delete the existing fingerprint. Wait for the Status to update to **deleted**.

- 2 In the same row, change the Command back to **register**, clear the Status cell, and clear the FingerID cell.
- 3 Have the employee stand at the device and follow the enrollment steps in Chapter 4 from Step 4 onwards.

6.4 Changing the Office WiFi Network

If your office WiFi name or password changes, the device will lose its connection and the LED will glow steady **purple**. You will need the master finger to reconfigure it.

- 1 Make sure the device is powered on.
- 2 Place the master finger on the sensor. The LED will flash yellow five times, then turn purple — the WiFi setup portal is now running.
- 3 On your phone or laptop, connect to the **Attendance-Setup** WiFi network (password: setup1234).
- 4 Open a browser and go to **http://192.168.4.1**.
- 5 Enter the new WiFi name and password, then click **Save & Connect**.
- 6 The device will restart and the LED should return to steady blue.

6.5 Emergency Portal Access (No Master Finger)

If the master finger is unavailable — lost, injured, or not yet enrolled — you can still trigger the WiFi setup portal using a built-in software fallback. This requires no hardware tools.

NOTE

This fallback exists precisely for emergencies. You should still enrol a master finger as soon as possible. The fallback should not be relied upon as the primary method.

How to trigger the emergency portal:

- 1 Make sure the device is powered on and the LED is **purple** (no WiFi).
- 2 Press any **unregistered** finger on the sensor — one that is not enrolled in the system. The LED will flash red briefly.
- 3 Lift the finger, wait for the red flash to end, then press again. Repeat this **5 times within 30 seconds**.
- 4 After the 5th no-match scan, the device will flash yellow three times and launch the WiFi portal automatically.
- 5 From this point, follow steps 3–6 in Section 6.4 to enter the new WiFi credentials.

WARNING

The finger must genuinely not match any enrolled fingerprint — using an enrolled employee's finger will simply record attendance, not count as a fallback tap. Use a finger that has never been enrolled, such as a little finger or a finger from a visitor.

6.6 Reviewing the Monthly Email Report

On the first day of each month, an email is automatically sent to all configured admin addresses with the previous month's attendance summary. The email contains a section for each active branch, listing:

- Employees with absences, and how many days they were absent.
- Employees with late arrivals, and how many times.
- Employees with early departures.
- Employees with missing check-outs.

The email also includes an Excel (.xlsx) attachment for each branch, named in the format **Branch Name - Attendance - Month Year.xlsx**. The attachment contains the full raw attendance log for that month and a Monthly Summary tab with aggregated counts per employee — useful for payroll or HR reviews.

TIP

If you do not receive the monthly email, check your spam or junk folder first. If it is not there, contact your system administrator.

6.7 Understanding the Monthly Archive

When the monthly report runs, the system automatically archives the previous month's raw attendance rows. Here is what happens:

- The Excel attachment is generated **first**, capturing the full raw data for that month.
- A new tab named **Archive - Month Year** (e.g. Archive - March 2026) is created inside your branch spreadsheet.
- All rows from the previous month are copied into that archive tab.
- Those rows are then removed from the main **Attendance** sheet, keeping it lean and fast.
- Any rows from the current month that may already be in the sheet are **not** touched.

NOTE

Historical data is never deleted — it is always preserved in the archive tabs inside the same spreadsheet. If you need to look up a past record, click the relevant archive tab at the bottom of the spreadsheet.

6.8 Correcting a Wrong Record

If a record needs to be corrected — for example a Time-In value is wrong due to a clock error, or an employee's name was misspelled — you can edit the Attendance sheet directly.

- Click the cell you need to change and type the correct value.
- For Time-In and Time-Out, use the format **HH:MM** (e.g. 08:15, 17:03).
- For Date, use the format **YYYY-MM-DD** (e.g. 2026-04-08).

- The system does not re-validate manually edited cells — whatever you type is saved as-is.

6.9 Checking for Missed Check-Outs

The nightly job at 11:30 PM automatically fills in **No Check-Out** for any employee who scanned in but never scanned out. If you want to check earlier in the day, open the Attendance sheet and look for rows where the **Time-Out** column is blank and the **Date** is today — those employees have arrived but not yet scanned out.

NOTE

Offline scans are queued on the device and sent when WiFi reconnects. If an employee scanned while the device was offline and a queued scan arrives after an Absent row was already written by the nightly job, the system will automatically correct the Absent row to a proper Time-In record. No manual action is needed.

Troubleshooting

Common problems and how to resolve them

7.1 Device LED Colour Guide

LED State	Meaning	Action
Steady blue	Connected to WiFi, ready.	None — normal operation.
Steady purple	No WiFi connection.	Reconfigure WiFi using master finger (Section 6.4) or emergency fallback (Section 6.5).
Yellow flash x5	Master finger detected — entering WiFi setup mode.	Proceed with setup if intended; otherwise unplug and replug to restart.
Yellow flash x3	Emergency portal fallback triggered.	Device is entering portal mode. Follow Section 6.5 from Step 5.
Solid yellow	Waiting for fingerprint during enrollment.	Place the correct finger on the sensor.
Green flash	Scan accepted / enrollment succeeded.	None.
Red flash	Finger not recognised, or enrollment failed.	Try again. If persistent, re-enrol (Section 6.3).

7.2 Common Problems and Solutions

Scans are not appearing in the Attendance sheet

- Check the device LED is steady **blue**. If it is purple, the device has no WiFi — reconfigure using the master finger (Section 6.4) or emergency fallback (Section 6.5).
- The device queues scans offline and sends them once WiFi is restored. After reconnecting, wait a minute and check if queued scans appear.
- If the LED is blue but records still do not appear, contact your system administrator with the branch name and the time the scan was made.

An employee's finger is not being recognised

- Ask the employee to wipe the sensor surface with a clean dry cloth, then try again.
- Ensure the employee places the **centre** of their fingerprint on the sensor — not the tip of the finger.
- If the problem continues after several attempts, delete and re-enrol the fingerprint (Section 6.3).

Enrollment is stuck — the Status column never updates from 'processing'

- The device waits up to 20 seconds for a finger during enrollment. If no finger was placed in time it times out. To retry: clear the Status cell and change the Command back to **register**.
- Confirm the device LED is steady blue (online) before retrying.
- Check the **Note** column — the device may have written an error code there.

The device is not picking up the enrollment command

- The device polls for new commands every 10 seconds. Wait at least 15 seconds after adding the row.
- Confirm the **Command** cell contains exactly **register**, **delete**, or **clearall** — no capital letters, no extra spaces.
- Confirm the **Status** cell is blank or contains **pending**. If it says **processing**, the device has already picked it up — just wait.
- Confirm the device LED is steady blue.

The master finger is triggering the WiFi portal unexpectedly

- This is normal behaviour — placing the master finger always launches the portal. To cancel, unplug and replug the device to restart it.
- If this keeps happening accidentally, consider enrolling the master finger using a less-used finger (e.g. little finger) to avoid accidental triggers.

The emergency portal fallback is not triggering

- Make sure you are using a finger that is **genuinely not enrolled** — you need to see a red flash on each tap for it to count.
- All 5 taps must happen within 30 seconds. If you take longer, the count resets from zero.
- After the 5th tap, watch for three yellow flashes — that confirms the portal is launching.

An employee was marked Absent but they did scan in (offline scan)

- This can happen if the device was offline when the employee scanned, and the nightly job ran before WiFi was restored.
- The system automatically corrects this: when the queued scan finally uploads, it will overwrite the Absent row with the correct Time-In. No manual action is needed.
- If the correction does not appear within a few minutes of WiFi reconnecting, contact your system administrator.

The monthly email was not received

- Check your spam or junk folder.
- Confirm your email address was provided to your system administrator to be added to the recipient list.
- If neither helps, contact your system administrator.

An archive tab is missing for a past month

- Archive tabs are created automatically when the monthly report runs on the first of each month.
- If a tab is missing, check the Excel attachment from the monthly email — it contains the full raw data for that month.
- Contact your system administrator if neither the tab nor the email attachment can be found.

If a problem is not covered in this guide, contact your system administrator and provide: the branch name, a description of the issue, the LED colour or state at the time, and — if relevant — the exact text shown in the Status and Note columns of the Enrollment sheet.